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Primary Kangaroo

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Solutions

These are polished solutions and do not illustrate the process of failed ideas and rough work by which candidates may arrive at their own solutions.

It is not intended that these solutions should be thought of as the 'best' possible solutions and the ideas of readers may be equally meritorious.

Enquiries about the Primary Kangaroo should be sent to:

challenges@ukmt.org.uk

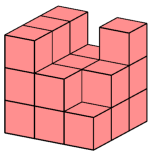
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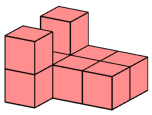
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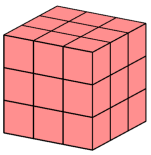
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A	E	E	C	A	B	A	E	B	E	D	C	D	B	B	E	D	D	C	C

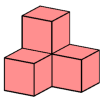
1. Mia is adding small cubes one by one to build a $3 \times 3 \times 3$ cube.
She took the five photographs shown below at five different times.

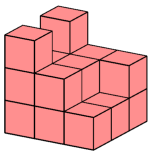
What does the fourth photograph that Mia took look like?

A 

B 

C 

D 

E 

SOLUTION **A**

As Mia builds the $3 \times 3 \times 3$ cube, the number of the small cubes she uses increases. So the order of her photos is D, B, E, A and finally C. So her fourth picture looks like A.

2. Simon writes the four digits 2, 0, 2, 5 in the four boxes below.

$\square + \square - \square + \square$

Which order would give him the largest result?

A 0, 2, 2, 5 B 0, 5, 2, 2 C 2, 5, 2, 0 D 5, 0, 2, 2 E 5, 2, 0, 2

SOLUTION **E**

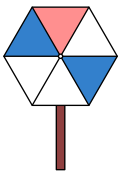
Each of the options includes the same group of digits, but in different orders. The third digit is subtracted, while the other three digits are added. Therefore the result is largest when the third digit is the smallest, that is 0. The third digit is 0 only in option E.
It is, of course, also possible simply to calculate all five results and find the largest of them:

A: $0 + 2 - 2 + 5 = 5$ B: $0 + 5 - 2 + 2 = 5$ C: $2 + 5 - 2 + 0 = 5$

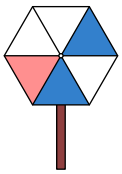
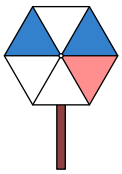
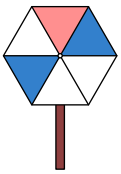
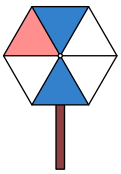
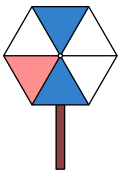
D: $5 + 0 - 2 + 2 = 5$ E: $5 + 2 - 0 + 2 = 9$

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3. Larissa has a pinwheel, which is shown below.



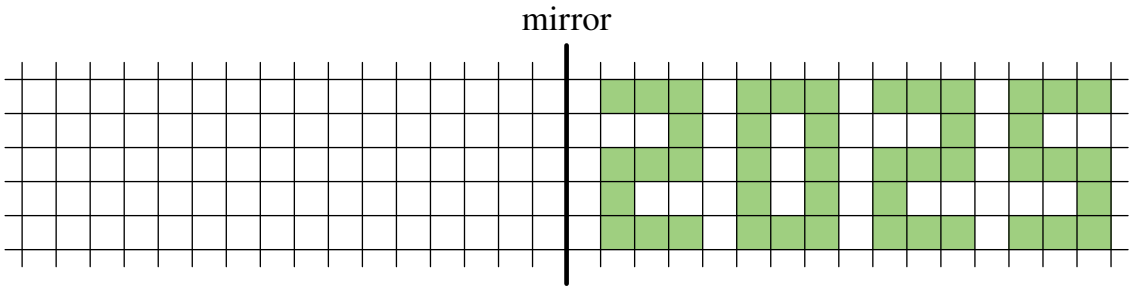
She spins her pinwheel. Which of the pinwheels shown below could be hers?

A 
B 
C 
D 
E 

SOLUTION **E**

On Larissa's pinwheel, the two blue triangles are directly opposite each other. This is not true for the wheels in options A and B. The pink triangle in Larissa's wheel is immediately next to a blue triangle in a clockwise direction. Of the three options remaining, C, D and E, this is true only in option E.

4. Otto reflects the number 2025 in the mirror on its left.

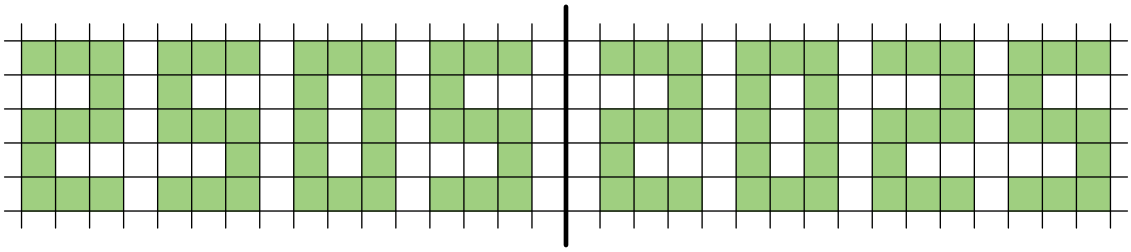


What is the difference between 2025 and its reflection?

A 350 B 477 C 480 D 2840 E 2997

SOLUTION **C**

The reflection of 2025 in the mirror looks as shown below:



Hence the difference between 2025 and its reflection is $2505 - 2025 = 480$.

5. On a normal dice, the total number of spots on two opposite faces is always 7.

Only one of the dice shown below could be a normal dice. Which one?

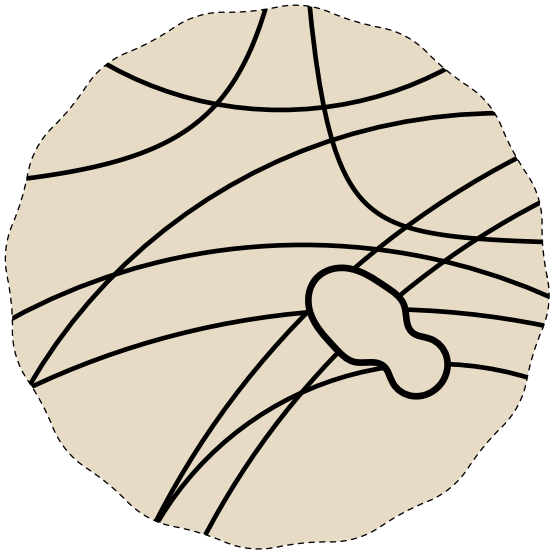
A  B  C  D  E 

SOLUTION **A**

On a normal dice the total number of spots on two opposite faces is always 7. So the numbers of dots on opposite faces are, in pairs, 1 and 6, 2 and 5, and 3 and 4. In options B and E, 2 dots and 5 dots are on neighbouring faces (and so not on opposite faces); in option C, 1 dot and 6 dots are on neighbouring faces, as are the 3 dots and 4 dots in option D. The only option left is A, which *could* be a normal dice. (Of course, it still might not be!)

6. Alice stepped on some tracks on the ground.

Which track pattern was under her foot?



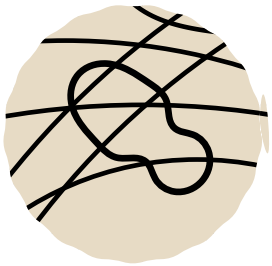
A  B 

C  D 

E 

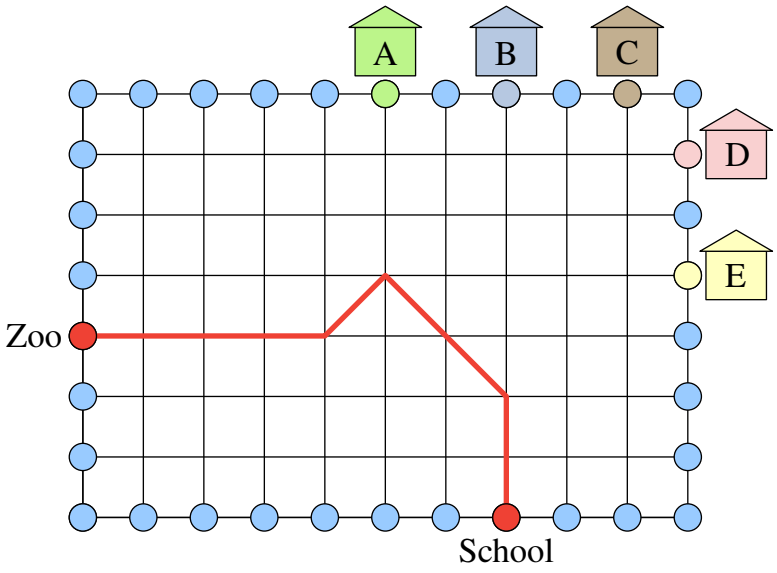
SOLUTION **B**

If we connect the tracks as shown on the right, we can see that option B has the correct track pattern.



[You might like to see if you can find where the foot would have been for each of the other options.]

7. Kangaroo jumps from School to the Zoo, as follows: ↑ 2, ↖ 2, ↙ 1, ← 4, as shown below:



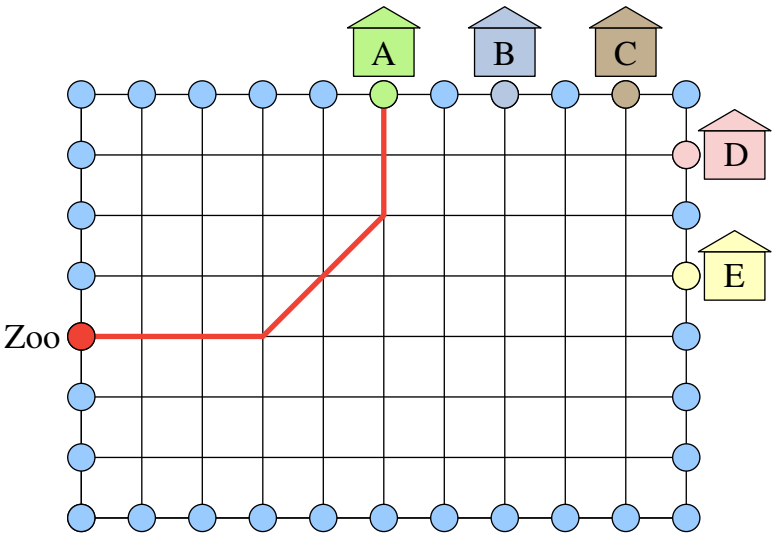
Now he jumps from the Zoo, as follows: → 3, ↗ 2, ↑ 2.

Which house will he arrive at?

A  B  C  D  E 

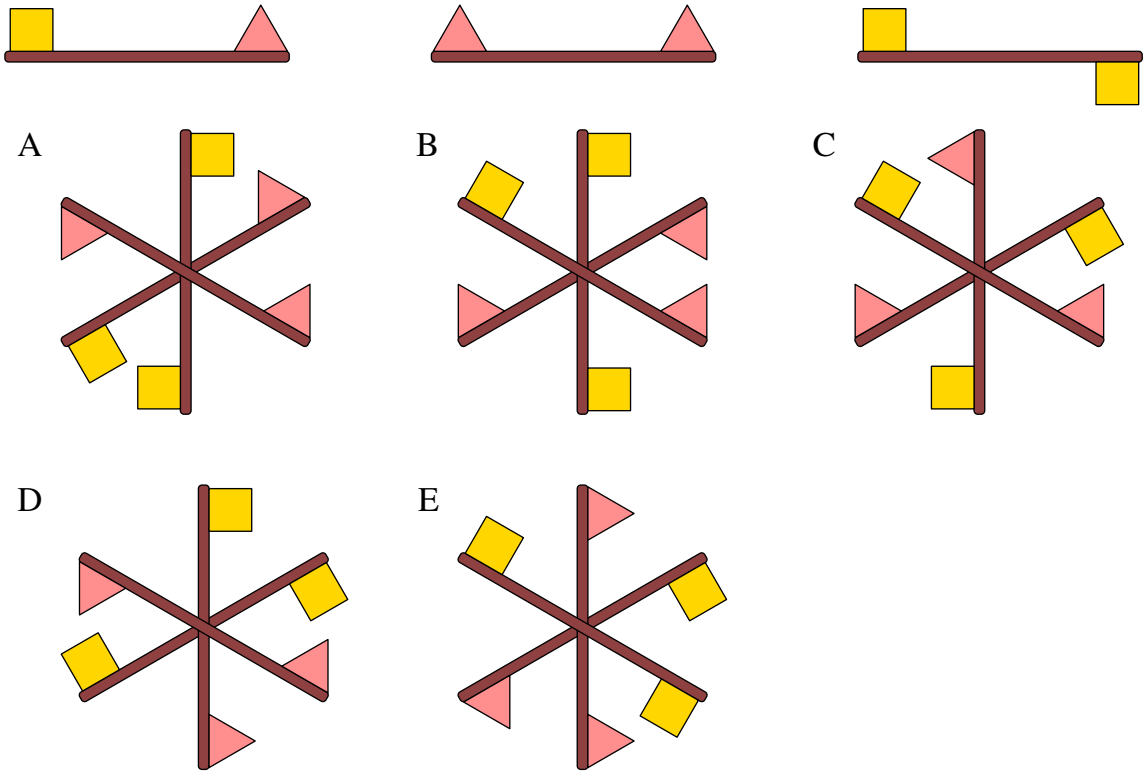
SOLUTION **A**

We start at the Zoo and draw the path that Kangaroo covers according to the instructions:



Kangaroo arrives at house  .

8. Which pinwheel can Jorge build with these three rods?

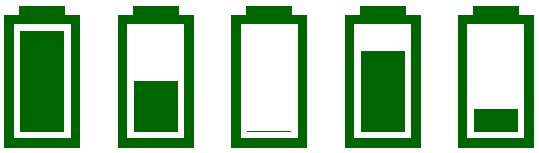


SOLUTION **E**

There are many ways to approach this, but one is to focus on the second rod, which has two pink triangles on the same side. In each of options A, B and D, the rod with two triangles has them on different sides. The wheel in option C cannot be made using the three rods shown, as it does not use the second rod but instead requires a rod with a square and a triangle on different sides. Only option E has the two triangles on the same side, and it is easy to see how the other two rods fit to make that wheel.

9. In the morning, five friends (Ann, Bob, Cristina, David and Edward) had the same 'fully-charged' mobile phones. By the evening, Bob had spoken on the phone as much as Ann and Cristina together. Bob's phone ran out of power. David had not used his phone at all.

The battery levels of their phones are shown on the right.



Which phone belongs to Edward?

A 1 B 2 C 3 D 4 E 5

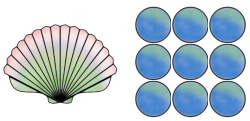
SOLUTION **B**

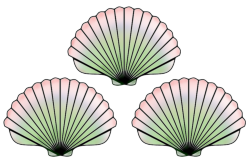
As Bob's phone ran out of power, his is phone 3. David has not used his phone at all, so his is phone 1. The battery usage for phones 4 and 5 together amounts to a whole battery, as used by Bob, so phones 4 and 5 belong to Ann and Cristina, in some order. Only phone 2 remains, so this belongs to Edward.

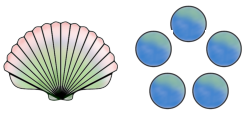
10. Nico and his little sister are playing with shells and marbles.
In their shop, each shell has a value of 6 and each marble has a value of 1.

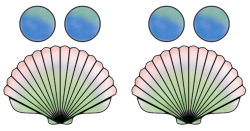
Which of the following collection of shells and marbles has a total value of 16?

A 

B 

C 

D 

E 

SOLUTION **E**

We calculate the value for each collection.
In collection A, the two shells have a value of $2 \times 6 = 12$, so the total value is $12 + 1 = 13$.
In B, the shell has a value of 6, so the total value is $6 + 9 = 15$.
In C, the three shells have a value of $3 \times 6 = 18$.
In D, the shell has a value of 6, so the total value is $6 + 5 = 11$.
In E, the two shells have a value of $2 \times 6 = 12$, so the total value is $12 + 4 = 16$.
So only collection E has a total value of 16.

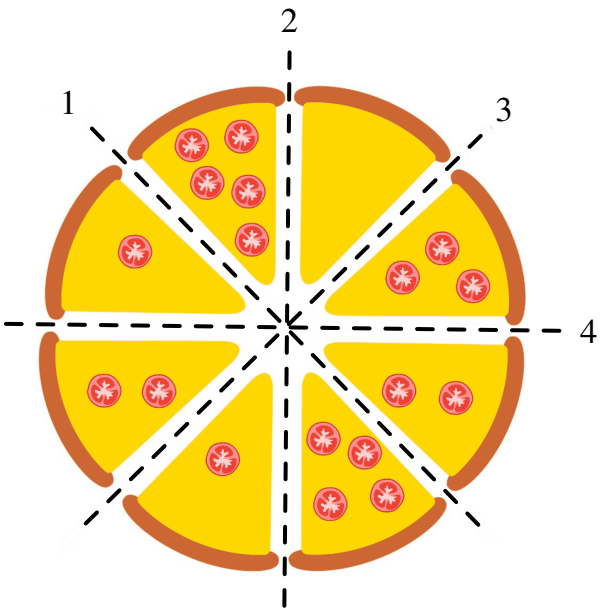
An alternative way, with far less calculation, is to notice that the total value of each of the collections A, B and D is an odd number and the total value of collection C is a multiple of 3. So only collection E could have a total value of 16, which indeed it does.

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11. Tom wishes to slice a pizza into 2 halves. He also wishes to have the same number of tomatoes on each half. It is possible for him to do this with two different cuts.

Along which lines could he cut?

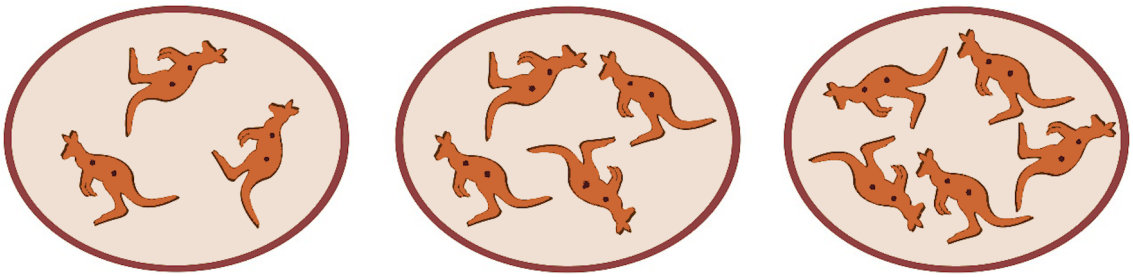
A 1 and 3 B 1 and 4
C 2 and 3 D 2 and 4
E 3 and 4



SOLUTION **D**

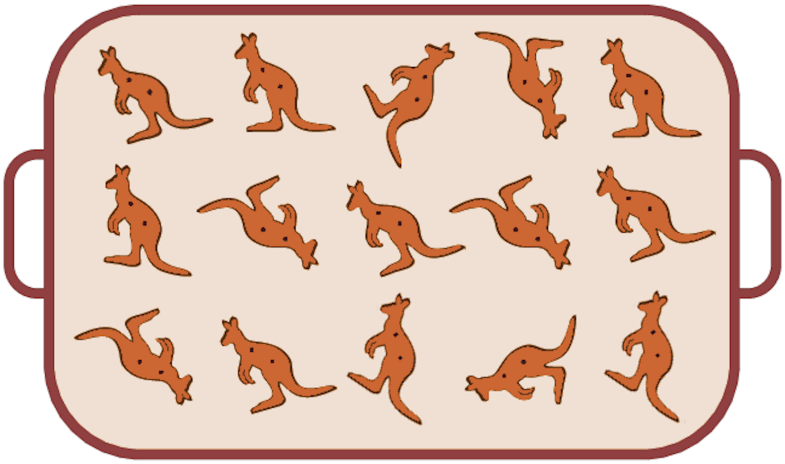
We start with cut 1 and count the number of tomatoes on either side. The number of tomatoes above cut 1 is $5+0+3+2=10$ and the number of tomatoes below cut 1 is $4+1+2+1=8$. Similarly, the number of tomatoes to the left of cut 2 is $5+1+2+1=9$ and the number of tomatoes to the right is $0+3+2+4=9$. The number of tomatoes above cut 3 is $0+5+1+2=8$ and the number of tomatoes below cut 3 is $1+4+2+3=10$. Finally, the number of tomatoes above cut 4 is $1+5+0+3=9$ and the number of tomatoes below cut 4 is $2+1+4+2=9$. Therefore to have the same number of tomatoes on each half of the pizza, Tom could cut along line 2 or line 4.

12. Anna, Bonnie and Caspar already have some kangaroo cookies on their plates.



Anna Bonnie Caspar

They share the remaining 15 cookies on the tray shown below so that everyone will have the **same** number of cookies on their plates.



How many *more* cookies must Anna receive?

A 4 B 5 C 6 D 7 E 8

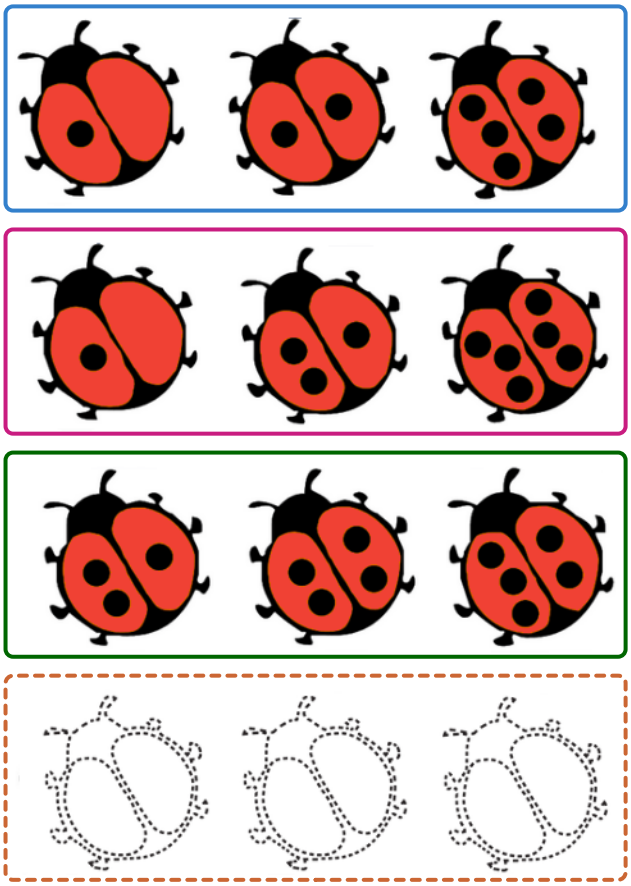
SOLUTION **C**

The numbers of cookies Anna, Bonnie and Caspar have already are 3, 4 and 5 respectively. If we put two more cookies on Anna's plate and one more cookie on Bonnie's plate, then they will all have the number of cookies on their plates. The number of cookies still to be shared is $15 - 2 - 1 = 12$ and, since these should be divided equally, the number of cookies each child will get from the 12 cookies remaining is $12 \div 3 = 4$. Therefore the number of extra cookies Anna should receive is $2 + 4 = 6$.

Another solution is first to calculate how many cookies each child receives in total. They already have 3, 4 and 5 cookies. Along with these 15 cookies, there are $3 + 4 + 5 + 15 = 27$ cookies in total. So, at the end each child should have $27 \div 3 = 9$ cookies. As a result, Anna must get $9 - 3 = 6$ more cookies.

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13. Six ladybirds have 1, 2, 3, 4, 5 or 6 spots each. Marta took four photos of them in groups of 3. Each ladybird appeared the same number of times in the four photos. Three of the photos are shown here, along with the outline of the fourth photo.



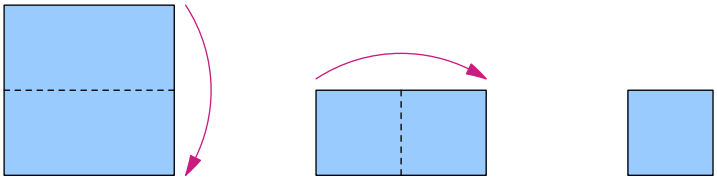
How many spots do the three ladybirds in Marta's fourth photo have in total?

A 9 B 10 C 11 D 12 E 23

SOLUTION **D**

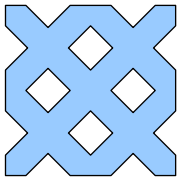
Each of the ladybirds with 1, 3 and 5 spots appears twice in the first three photos, and those with 2, 4 and 6 appear only once. So the fourth photo will show three ladybirds, one with 2 spots, one with 4 spots and one with 6 spots. Hence the total number of spots on the three ladybirds in the fourth photo is $2 + 4 + 6 = 12$.

14. Nela folds a paper square in half and then in half again, as shown.



Next she cuts pieces out of the folded paper.

After unfolding, she sees a paper snowflake.

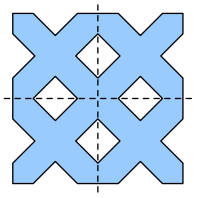


How did she cut the folded paper?

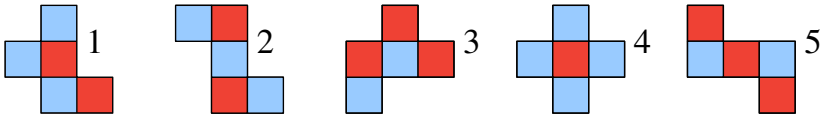
A  B  C  D  E 

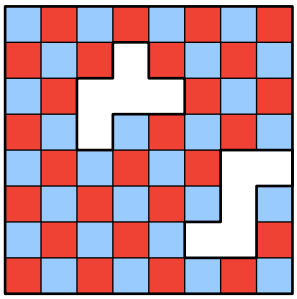
Solution **B**

If the two folding lines are added to the snowflake as shown on the right, one can see that each quarter resembles the pattern shown in option B. This must have been this cutting pattern that Nela used.



15. Which two of the pieces below complete the chessboard?



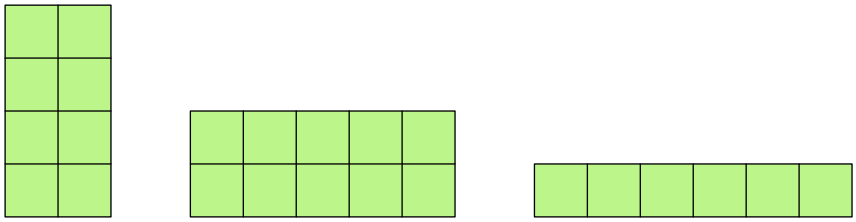


A pieces 1 and 2
B pieces 1 and 5
C pieces 3 and 4
D pieces 3 and 5
E pieces 4 and 5

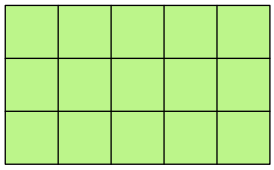
Solution **B**

t is impossible for piece 4 to fit in either of the holes. If we turn piece 1 around so that three of the squares are in a horizontal row, we can see that both piece 1 and piece 3 would fit in the upper hole. However, only piece 1 has the correct colouring. Similarly, if we turn piece 5 so that three of the squares are in a vertical column, we see that it fits the hole on the right, and that its colouring is correct. There is no way to fit piece 2 into the same hole without reflecting it, and even then its colouring would be incorrect. So pieces 1 and 5 will complete the chessboard.

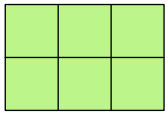
16. Bob makes a square from four rectangular pieces. Three of the pieces are shown below.

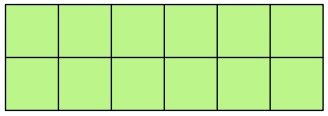


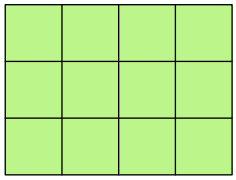
Which one is the fourth piece?

A 

B 

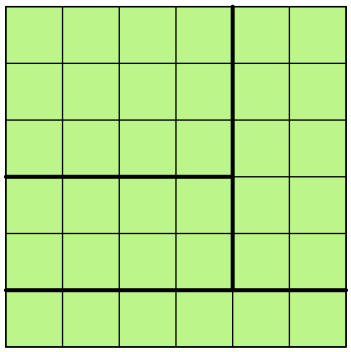
C 

D 

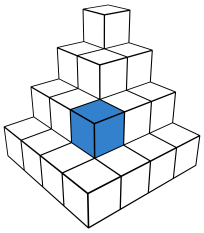
E 

SOLUTION **E**

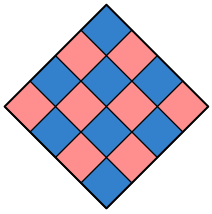
The largest side-length of one of the three given pieces is 6. Therefore Bob's large square will have a side-length of at least 6. Hence the total number of squares in Bob's large square is at least $6 \times 6 = 36$. The number of squares in the three given pieces is $8 + 10 + 6 = 24$. Therefore the number of squares in the fourth piece is at least $36 - 24 = 12$. Both rectangle D and rectangle E contain 12 squares and, as the diagram on the right shows, it is rectangle E, which is four squares wide and three squares high, that will fit with the three given pieces to form a square.

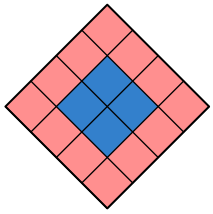


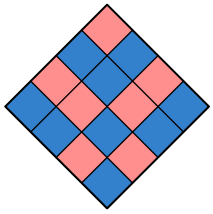
17. Léonie stacks up some pink and blue cubes into a pyramid as shown on the right.
One of the blue cubes is shown.
She arranges each cube so that each of its faces does **not** touch a face of another cube that has the same colour.

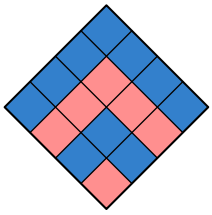


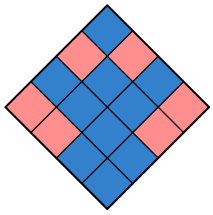
What will Léonie's pyramid look like from above?

A 

B 

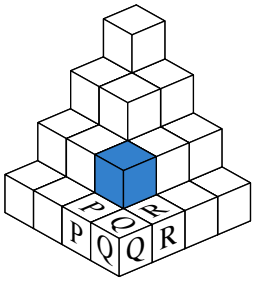
C 

D 

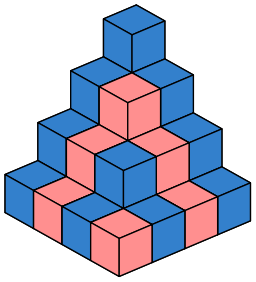
E 

SOLUTION **D**

One way to start is by thinking about the colour of the cube directly underneath the blue cube — it must be pink. This means that cubes P and R which touch that pink cube in the bottom layer must be both blue, and so (because it touches both P and R) cube Q must be pink.



You can now work your way along rows and columns, colouring alternately blue and pink to get the pyramid stack shown here. It should now be evident from above that the correct option is D.



18. The picture shows the page for one month of a calendar without any of the dates.

The total of the dates for the two shaded days is 29.

On what day of the week does the first day of this month fall?

Mon	Tue	Wed	Thu	Fri	Sat	Sun

A Monday B Tuesday C Wednesday D Thursday
E Sunday

SOLUTION **D**

Counting the number of days from the first shaded square to the second, we see that the second date is 13 days later than the first. So we need two numbers that have a total of 29 and a difference of 13.

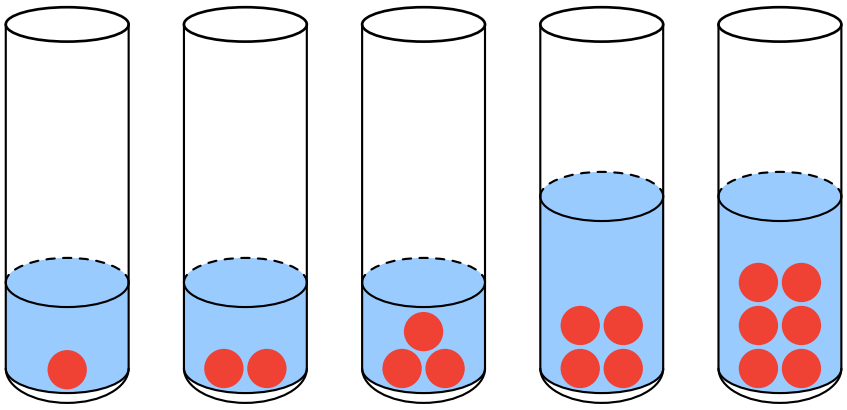
It doesn't take so long to find that these two numbers must be 8 and 21. The first day of the month is exactly one week before the eighth day, and since the eighth day is Thursday, so also is the first day of the month. (In fact, this is the calendar for May 2025).

Mon	Tue	Wed	Thu	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

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19. Identical balls were placed in five identical test tubes as shown.

Then water was added to each of these test tubes.



The water level in test tubes 1, 2, and 3 is the same.
The water level in test tubes 4 and 5 is also the same but is twice as high as in the first three.

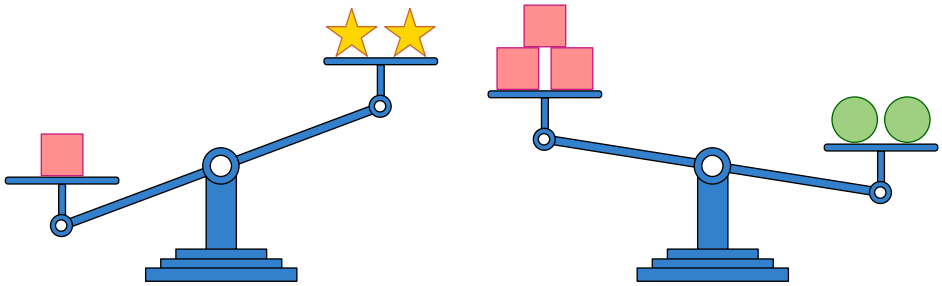
When all the balls are removed, which test tube will contain the *least* amount of water?

A test tube 1 B test tube 2 C test tube 3 D test tube 4 E test tube 5

SOLUTION **C**

The amount of water in tube 3 is less than the water in tube 1 and in tube 2 because, even though the water level is the same, tubes 1 and 2 contain a greater number of balls than tube 3. For the same reason, the amount of water in tube 5 is less than that in 4. Since we are looking for the tube which contains the least amount of water, all we must do now is to compare tube 3 with tube 5. Now imagine having another test tube exactly like 3 (with the same amount of water and the same three balls in each). Pour everything (water and balls) from these two tubes into a single test tube: we would produce a tube that looks identical to tube 5. Therefore, tube 5 contains not only twice as many balls as test tube 3 but also twice as much water. So, of the five test tubes, test tube 3 has the least amount of water.

20. A pair of scales is used to weigh 3 different objects, and the results are shown below.



Each kind of object (cube, star or ball) has a different mass. The masses can be 1, 2, 3, 4 or 5 kg.

What is the mass in kilograms of one cube?

A 1 B 2 C 3 D 4 E 5

SOLUTION **C**

The smallest mass that one star could have is 1 kg.
The first weighing shows that one cube is heavier than two stars. Because of this, one cube is heavier than 2 kg; in other words 3 kg, 4 kg or 5 kg.
The second weighing shows that the two balls are heavier than the three cubes. The mass of the three cubes is at least 9 kg and hence the mass of the two balls is larger than 9 kg. Since the masses of each of the three objects is 1 kg, 2 kg, 3 kg, 4 kg or 5 kg, the mass of one ball is 5 kg.
Therefore, since all the object have different masses, the mass of a cube is either 3 kg or 4 kg.
However, if the mass of the cube was 4 kg then three cubes would be heavier than two balls as $3 \times 4 > 2 \times 5$, which would contradict what is shown in the second picture.
Therefore, the mass of one cube is 3 kg (and the mass of a star is 1 kg).

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